Bihar Engineering University, Patna
End semester Examination - 2023

Course: B.Tech.	*Semester-VI*	*Time: 03 Hours*
Code: 105610	*Subject: Soft Computing*	*Full Marks: 70*

Instructions:-
(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Define the following terms (any seven): **[2 x 7 = 14]**

(a) Bayesian Network
(b) Soft computing
(c) Feed forward network
(d) Reinforcement learning
(e) Fuzzy inference system
(f) Features of genetic algorithms (GA)
(g) Bayes classifier
(h) Perceptron model
(i) Fuzzy operators
(j) Application of artificial neural network

Q.2 (a) What are different types of neural networks based on architecture? Explain. **[7]**
(b) Describe backpropagation algorithm. What are advantages and disadvantages of backpropagation algorithm. **[7]**

Q.3 (a) What is genetic algorithm? Explain with flow chart. **[7]**
(b) Describe various operators of Genetic Algorithm. **[7]**

Q.4 (a) What tasks can be performed with feedback NN? Explain Hopfield net. **[7]**
(b) Define fuzzification and explain any two methods of it **[7]**

Q.5 (a) What do you understand by soft computing? How is it related to machine learning? **[7]**
(b) Explain different techniques used in soft computing. **[7]**

Q.6 Consider two fuzzy sets **[4+4+6=14]**

$$A = \left\{\frac{0.2}{0} + \frac{0.3}{1} + \frac{1}{2} + \frac{0.1}{3} + \frac{0.5}{4}\right\}, \qquad B = \left\{\frac{0.1}{0} + \frac{0.25}{1} + \frac{0.9}{2} + \frac{0.7}{3} + \frac{0.3}{4} + \frac{0.2}{5}\right\}$$

Find the following:
(a) Algebraic sum (b) Bounded sum (c) Bounded Difference

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Q.7 What is the effect of Gaussian environment on perceptron and Bayes classifier? Develop the relationship between them in Gaussian environment. **[14]**

Q.8 (a) Write a python program to find largest of three numbers. **[7]**
(b) Given two universes X = { x1, x2, x3, x4, x5} and Y = {y1, y2, y3, y4, y5}, the fuzzy sets A defined on X and fuzzy set B defined on Y are given below. **[7]**
Find the relation R = A x B, where

$$A = \left\{\frac{0.4}{x1} + \frac{0.7}{x2} + \frac{1}{x3} + \frac{0.8}{x4} + \frac{0.6}{x5}\right\}, \qquad B = \left\{\frac{0.2}{y1} + \frac{0.6}{y2} + \frac{1}{y3} + \frac{0.9}{y4} + \frac{0.7}{y5}\right\}$$

Q.9 Write short notes on ***any two*** of the following: **[7x2=14]**
(i) Soft versus hard computing
(ii) Fuzzy relation
(iii) Application of GA in machine learning

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